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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
Weekly Progress Report (WPR)

WPR Dates: 30/1/2026 - 6/2/2026 **WPR of Week:**3

Name: Santosh Kumar

Enrollment Number: 12196502722

Class & Section: CSE-8A

Department: Computer Science and Engineering

Faculty Supervisor's Name: Ms. Indu Sharma

MAJOR PROJECT TITLE: - "AURA : Autonomous User Realization & Assembly"

TARGETS SET FOR THE WEEK:

- Design the canonical JSON structure for deterministic circuit representation.
- List supported component metadata fields needed for pins, constraints, and inventory linkage.
- Identify the first local calculation and rule-checking cases for supported circuit patterns.
- Plan how AI-assisted circuit output will remain structured and reviewable.
- Define the first bridge points between inventory records and circuit recommendations.

ACHIEVEMENTS FOR THE WEEK:

- Prepared the first draft of the canonical JSON model for structured circuit state.
- Defined key metadata fields for pins, package constraints, and inventory association.
- Selected initial deterministic checks for simple circuit reasoning and supported-pattern analysis.
- Documented the requirement that AI suggestions must remain structured rather than free-form.
- Outlined the first connection points between owned inventory and circuit recommendation logic.

FUTURE WORK PLANS:

- Refine the workspace flow for viewing and editing structured circuit data.
- Begin planning the local-first editing model before backend synchronization.
- Prepare the first concept for proposal compare and review behavior.
- Continue narrowing the supported V1 feature set for deterministic operation.

Week's Summary

Days/Time	5 Days a week Monday to Friday.
Monday	Drafted the canonical JSON structure for deterministic circuit representation.
Tuesday	Defined metadata fields for pins, package information, and component constraints.
Wednesday	Selected initial local rule-checking and supported-pattern cases for early analysis.
Thursday	Outlined how AI-assisted circuit suggestions should remain structured and reviewable.
Friday	Connected inventory constraints with the first circuit recommendation assumptions.

STATUS OF RESEARCH PAPER: Background notes were extended to deterministic circuit modeling, structured AI output, and constraint-driven workflows. Estimated overall project completion by the end of this week: 28%.

Supervisor signature: **Date:**